Extremizers for the independent Berry–Esseen inequality

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Abstract

We study the variational structure of distributions maximizing the Berry–Esseen ratio for independent centered real random variables. First, at any fixed-n or compactified maximizer, a second-order probability variation preserving mass and mean proves that every summand is supported on at most four points; at a global maximum, any four-point summand lies on a flat probability segment whose endpoint replaces it by a three-point summand with the same variance and third absolute moment. By the strict Cox–Kemperman third-moment inequality $\mathbb{E}|U+W|^3 > \mathbb{E}|U|^3 + \mathbb{E}|W|^3$ for nonconstant independent centered variables, every summand of an unrestricted maximizer is convolution-indecomposable. Second, if the universal independent Berry–Esseen constant C_{ind} exceeds Esseen's constant $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$, the supremum is attained in a Lévy–Khintchine compactification consisting of countably many independent summands plus an infinitely divisible remainder. For any three-point maximizing representation, we prove that both the Gaussian and Lévy jump components vanish, the summand support diameters tend to zero, and any infinitely active sum is nonatomic. Under the additional third-moment bound $\beta \le 18481/25000$ at unit variance, multi-scale grouping and randomized lattice rounding establish weighted variance summability $\sum_i v_i \operatorname{diam}(\operatorname{supp} X_i)^{-\gamma} < \infty$ for all $\gamma < \sqrt{5}-1$ (and all $\gamma < \sqrt{21}-3$ when all but finitely many summands are two-point), ruling out two-sided $C^{1,1}$ regularity of the distribution function at a maximizing threshold as well as separated finite blocks and dense or nonoverlapping rare Bernoulli tails.

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1 Introduction and main results

1.1 Formulation of the problem

For a centered real random variable S of variance one, write

$$\Delta(S) = \sup_{x\in\mathbb{R}} |\mathbb{P}(S \le x) - \Phi(x)|,$$

where Φ and ϕ denote the standard normal distribution function and density. The universal Berry–Esseen constant for independent summands is

$$C_{\mathrm{ind}} = \sup \frac{\Delta(\sum_{i=1}^n X_i)}{\sum_{i=1}^n \mathbb{E}|X_i|^3}, \qquad C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}} = 0.4097321837\ldots,$$

where the supremum is taken over all finite independent centered arrays $(X_1, \ldots, X_n)$ with $\sum_{i=1}^n \mathbb{E}X_i^2 = 1$ and $\sum_{i=1}^n \mathbb{E}|X_i|^3 < \infty$. Esseen's two-point construction gives $C_{\mathrm{ind}} \ge C_{\mathrm{E}}$ [6, 14].

A companion paper [3, Theorem 1.1], extending the method of He and Cheng [7] to non-identically distributed summands via the common-diameter theorem of Bell [2], establishes the sharp bound in the small-Lyapunov regime: with $\ell_0 = \exp(-\exp(97))$,

$$\sum_{i=1}^n \mathbb{E}|X_i|^3 \le \ell_0 \quad \Longrightarrow \quad \Delta\left(\sum_{i=1}^n X_i\right) \le C_{\mathrm{E}} \sum_{i=1}^n \mathbb{E}|X_i|^3. \tag{1.1}$$

Consequently, any sequence of unit-variance arrays whose Berry–Esseen ratios exceed C_{E} must have third-moment sums bounded below by $\ell_0 > 0$. This uniform lower bound prevents maximizing sequences from degenerating to a pure Gaussian limit and allows us to study the extremal problem at positive Lyapunov ratio via compactification and variational calculus.

1.2 Compactification and structural theorems

When passing to a limit of finite independent arrays $(X_{n,1}, \dots, X_{n,k_n})$ ordered by decreasing variance, individual summands whose variances remain bounded away from zero converge weakly to countable summands X_i, while vanishingly small summands can coalesce into an independent infinitely divisible remainder Y by the classical accompanying-Poisson limit theorem [11].

Definition 1.1. An *admissible representation* is a decomposition

$$S = \sum_{i\geq 1} X_i + Y,$$

$$\log \mathbb{E}e^{\mathrm{i}tY} = -\frac{at^2}{2} + \int_{\mathbb{R}\setminus\{0\}} (e^{\mathrm{i}ty} - 1 - \mathrm{i}ty)\,\nu(dy), \tag{1.2}$$

$$\sum_{i\geq 1} \mathrm{Var}(X_i) + a + \int_{\mathbb{R}\setminus\{0\}} y^2\,\nu(dy) = 1, \qquad \beta = \sum_{i\geq 1} \mathbb{E}|X_i|^3 + \int_{\mathbb{R}\setminus\{0\}} |y|^3\,\nu(dy) < \infty, \tag{1.3}$$

where $(X_i)_{i\geq 1}$ are independent centered random variables, Y is a centered infinitely divisible random variable independent of $(X_i)_{i\geq 1}$, $a \geq 0$ is the Gaussian variance, and ν is a Lévy measure on $\mathbb{R}\setminus\{0\}$. We call β the *additive third-moment functional* of the representation. An admissible representation is *maximizing* if $\beta > 0$ and $\Delta(S) = C_{\mathrm{ind}}\beta$.

Note that the Gaussian variance a contributes to the unit-variance constraint in (1.3) but contributes zero to β, because a normal variable $N(0, a)$ is a weak limit of sums of N symmetric two-point variables of total variance a whose third-moment sums $a^{3/2}/\sqrt{N}$ tend to zero (Lemma 2.3). Thus β depends on the representation (X_i, a, ν) rather than on the marginal law of S alone.

Our first main theorem establishes the existence of maximizing representations when $C_{\mathrm{ind}} > C_{\mathrm{E}}$, proves the unconditional four-point and three-point support restrictions (stated for both finite-n and compactified maximizers in Theorem 3.2 below), and eliminates both the Gaussian and Lévy jump components.

Theorem 1.2 (Support reduction and elimination of infinitely divisible components)**.** *If $C_{\mathrm{ind}} > C_{\mathrm{E}}$, there exists a maximizing admissible representation. Every maximizing representation has at most four support points per discrete summand X_i, and admits an equally maximizing representation with at most three support points per summand that preserves each individual variance $\mathrm{Var}(X_i)$, each third absolute moment $\mathbb{E}|X_i|^3$, and the remainder Y. Moreover, every maximizing representation with at most three support points per summand satisfies*

$$a = 0, \qquad \nu = 0, \qquad \max\{|u| : u \in \operatorname{supp} X_i\} \longrightarrow 0 \quad (i \to \infty). \tag{1.4}$$

If infinitely many X_i are nonconstant, then the distribution of S is nonatomic and

$$\sum_{i\geq 1} (1 - \max_{u\in\mathbb{R}} \mathbb{P}(X_i = u)) = \infty.$$

After reflecting $S \mapsto -S$ if necessary, a maximizing representation attains its discrepancy on the positive closed branch at some threshold $z \in \mathbb{R}$:

$$F(z) - \Phi(z) = D = C\beta > 0, \qquad C = C_{\mathrm{ind}}, \qquad A = \frac{3D + z\phi(z)}{2}, \qquad F(x) = \mathbb{P}(S \leq x). \tag{1.5}$$

Our second main theorem establishes weighted variance summability and excludes smooth local expansions of F at z whenever β satisfies the upper bound $\beta \leq 18481/25000 = 0.73924$.

Theorem 1.3 (Weighted variance summability)**.** *Suppose $C_{\mathrm{ind}} > C_{\mathrm{E}}$, and let a maximizing representation with at most three support points per summand satisfy $\beta \le 18481/25000$. For each nonconstant summand X_i, set $H_i = \operatorname{diam}(\operatorname{supp} X_i)$ and $v_i = \operatorname{Var}(X_i)$. Then*

$$\sum_{i\ge 1} \frac{v_i}{H_i^{\gamma}} < \infty \quad (0 < \gamma < \sqrt{5} - 1), \qquad \sum_{i\ge 1} \mathbb{E}|X_i| < \infty.$$

If all but finitely many nonconstant summands are supported on two points, then $\sum_{i\ge 1} v_i H_i^{-\gamma} < \infty$ holds for the larger range $0 < \gamma < \sqrt{21} - 3$.

Corollary 1.4 (Excluded local regularity at a maximizing threshold)**.** *Under the hypotheses of Theorem 1.3, a maximizing threshold z satisfying (1.5) cannot satisfy the two-sided expansion*

$$F(z + t) = F(z) + \phi(z)t + O(t^2) \qquad (t \to 0).$$

If all but finitely many nonconstant summands have two support points, z cannot satisfy the one-sided left expansion

$$F(z) - F(z - h) = \phi(z)h + O(h^{1+\alpha}) \qquad (h \downarrow 0)$$

for any $\alpha > (\sqrt{21} - 3)/6$.

Our third main theorem restricts the combinatorial structure of support configurations that can reach the maximizing threshold z. For any finite index set I, put $T_I = S - \sum_{i\in I} X_i$ and define the *threshold completion set*

$$\mathcal{A}_I = \left\{ (x_i)_{i\in I} \in \prod_{i\in I} \operatorname{supp} X_i : \; z - \sum_{i\in I} x_i \in \operatorname{supp} T_I \right\}, \tag{1.6}$$

where $\operatorname{supp} T_I$ is the topological support of the independent remainder T_I.

Theorem 1.5 (Full affine rank of threshold completions)**.** *Fix a maximizing representation with at most three support points per summand as in Theorem 1.2, and a threshold z satisfying (1.5). There exists $k_0 \ge 1$ such that for every finite subset $I \subset \{k_0, k_0 + 1, \dots\}$, if functions $f_i : \operatorname{supp} X_i \to \mathbb{R}$ $(i \in I)$ satisfy*

$$\sum_{i\in I} f_i(x_i) = c \qquad \text{for all } (x_i)_{i\in I} \in \mathcal{A}_I,$$

then every f_i is constant on $\operatorname{supp} X_i$. Equivalently, fixing a base point $u_i^0 \in \operatorname{supp} X_i$ for each $i \in I$, the indicator vectors $(\mathbf{1}_{\{x_i = u\}})_{i\in I,\, u\in \operatorname{supp} X_i \setminus \{u_i^0\}}$ for $(x_i)_{i\in I} \in \mathcal{A}_I$ span an affine subspace of full dimension $\sum_{i\in I}(|\operatorname{supp} X_i| - 1)$.

Corollary 1.6 (Exclusion of separated finite blocks)**.** *A maximizing representation as in Theorem 1.5 cannot consist of a finite-support prefix followed by infinitely many nonconstant finite blocks of finite-support summands whose block diameters and minimum nonzero block-sum spacings*

$$H_k = \sum_{i \text{ in block } k} \operatorname{diam}(\operatorname{supp} X_i), \qquad \delta_k = \min_{u \ne v \in \operatorname{supp}(\sum_{i \text{ in block } k} X_i)} |u - v|$$

satisfy

$$\sum_k H_k < \infty, \qquad \delta_k \ge \sum_{l > k} H_l. \tag{1.7}$$

In Section 13 we further show that a maximizing representation cannot contain an infinite sequence of rare two-point summands $h_j(p_j - B_j)$ $(p_j \to 0)$ whose gap decrements satisfy $h_j - h_{j+1} = O(h_j^{\nu})$ with $\nu > 1 + (\sqrt{5} - 1)/2$ (Theorem 13.1) or whose gaps are nonoverlapping in the sense that $\sum_{k>j} h_k \le h_j$ and $h_{j+1} \ge c h_j$ (Theorem 13.2).

1.3 Open questions and overview of the proofs

The theorems above place stringent constraints on any hypothetical violation of $C_{\mathrm{ind}} = C_{\mathrm{E}}$: a maximizing representation must be a pure discrete sum $S = \sum_i X_i$ of indecomposable three-point (or two-point) summands with no Gaussian or Poisson part, shrinking supports, nonatomic sum law when infinite, full completion rank, and (for $\beta \le 18481/25000$) finite weighted variance moments $W_\gamma < \infty$ and rough distribution function at z. To close the conjecture $C_{\mathrm{ind}} = C_{\mathrm{E}}$ along this variational route, what remains is to rule out the coexistence of infinitely many active summands (or finitely many summands with unequal support diameters) in the window $\ell_0 < \beta < 1.321$.

Let us outline the methods used across the paper:

- **Compactification and variational calculus (Sections 2–3):** Section 2 establishes compactness of finite arrays in the space of admissible representations via accompanying Poisson laws [11]. Section 3 derives the first- and second-order variational conditions for probabilities and Lévy intensities, proves the unconditional four-point and three-point support reductions (Theorem 3.2) by analyzing the rank of $(1, u, u^2, |u|^3)$ in the stationary second logarithmic derivative, proves convolution indecomposability via the Cox–Kemperman inequality [4, 13], and bounds point masses of infinite sums via Sperner's theorem [15].

- **Elimination of Lévy jumps, nonshrinking supports, Gaussian components, and atoms (Sections 4–6):** Section 4 shows that any Lévy support point or nonshrinking sequence of summand supports $u_n \to x \neq 0$ generates an infinite arithmetic progression of contacts $Q(nx) = 0$ $(n \ge 1)$ by testing the 2×2 Hessian between the old probability/intensity direction and a single adjoined three-point summand $(1-(1+\varepsilon)t)\delta_0 + t\delta_x + \varepsilon t\delta_y$, contradicting boundedness of the contact set. Section 5 rules out $a > 0$ by combining a Stein–Ornstein–Uhlenbeck zero-bias bound ($a < 1/10$), a binomial Edgeworth lattice replacement ($\beta < \frac{17}{20}\sqrt{a}$), and a Fourier inversion bound using the Korolev–Shevtsova compound-Poisson estimate [10]. Section 6 rules out atoms in infinitely active sums by coupling a single pivotal mass transfer with a collective product-law variation across the tail.

- **Completion rank, weighted variance bootstrap, and rare tails (Sections 7–13):** Section 7 proves Theorem 1.5 by combining support translations that hold $\mathcal{A}_I$ fixed with probability tilts. Sections 8–12 prove Theorem 1.3 via a bootstrap iteration: under $\beta \le 18481/25000$, scalar curvature margins force every late three-point summand to have balanced extreme points $(H_i \mathbb{E}|X_i| \le K v_i)$. Combining Cramér moderate deviations for quadratic majorants, randomized lattice phase rounding, and compound-Poisson intensity variations converts any available weighted variance moment $W_s < \infty$ $(s > 1)$ into $W_\gamma < \infty$ for all $\gamma < T(s) = (3s+4)/(s+5)$ (whose fixed point is $\sqrt{5}-1$), or $\gamma < T_{\mathrm{B}}(s) = (3s+12)/(s+9)$ (whose fixed point is $\sqrt{21}-3$) for two-point tails. Section 13 applies Neumann-series inversion of $(qI + pT_h)$ to exclude dense and nonoverlapping rare Bernoulli tails.

2 Compactification and attainment above C_{E}

Lemma 2.1 (Moment-face three-point replacement)**.** *For every $c > 0$, any finite independent unit-variance array satisfying $\Delta(\sum_{i=1}^n X_i) > c \sum_{i=1}^n \mathbb{E}|X_i|^3$ can be replaced by one with the same number of summands and the same individual variances, each supported on at most three points, satisfying the same strict inequality.*

Proof. Fix the attaining threshold z on the positive closed branch (after reflection if needed). Holding all summands except X_i fixed, maximize $\int \{G_i(z-u) - c|u|^3\}\, P(du)$ over probability measures P

with $\int u\,dP = 0$ and $\int u^2\,dP = v_i$. By the extreme-point principle for three-constraint moment sets [2, 16], the maximum is attained at an extreme point supported on at most three atoms; replacing each summand successively preserves the strict inequality. □

Lemma 2.2 (Sequential compactness in admissible representations)**.** *Let $(X_{n,1}, \dots, X_{n,k_n})$ be finite independent centered arrays with total variance one and third-moment sums $\beta_n \le M$. After passing to a subsequence, their sums converge weakly to an admissible representation S satisfying*

$$\beta \le \liminf_{n\to\infty} \beta_n.$$

If each $X_{n,i}$ is supported on at most three points, then so is each limiting discrete summand X_i.

Proof. Pass to a subsequence along which β_n converges to its liminf, and order each row by decreasing variance $v_{n,1} \ge v_{n,2} \ge \cdots$ (padding with zeros). Since $v_{n,k} \le 1/k$ and $\mathbb{E}[X_{n,k}^2 \mathbf{1}_{\{|X_{n,k}|>L\}}] \le M/L$, a diagonal subsequence gives $X_{n,k} \Rightarrow X_k$ with $\mathbb{E}X_k = 0$, $v_k = \mathbb{E}X_k^2 = \lim_n v_{n,k}$, and $r_k = \mathbb{E}|X_k|^3 \le b_k := \lim_n \mathbb{E}|X_{n,k}|^3$. Choosing $K_n \to \infty$ slowly enough that $\sum_{k\le K_n} X_{n,k} \Rightarrow \sum_{k\ge 1} X_k$ and $\sum_{k\le K_n} |v_{n,k} - v_k| \to 0$, define the quadratic tail measures

$$\Lambda_n(E) = \sum_{k>K_n} \mathbb{E}\big[X_{n,k}^2 \mathbf{1}_{\{X_{n,k}\in E\}}\big],$$

whose total masses converge to $v_0 = 1 - \sum_{k\ge 1} v_k$ and whose tails outside $[-L, L]$ are at most M/L. Along a further subsequence, $\Lambda_n \Rightarrow \Lambda$ with $\Lambda(\mathbb{R}) = v_0$. Because $|f_{n,k}(t) - 1| \le v_{n,k}t^2/2$ and $\max_{k>K_n} v_{n,k} \le 1/(K_n+1) \to 0$, telescoping gives

$$\left| \prod_{k>K_n} f_{n,k}(t) - \exp\Big\{ \sum_{k>K_n} (f_{n,k}(t) - 1) \Big\} \right| \le \frac{t^4}{8} \sum_{k>K_n} v_{n,k}^2 \longrightarrow 0. \tag{2.1}$$

Since $\sum_{k>K_n}(f_{n,k}(t)-1) = \int q_t(y)\,\Lambda_n(dy)$ with the bounded continuous integrand $q_t(y) = (e^{ity} - 1 - ity)/y^2$ ($q_t(0) = -t^2/2$), the tail sums converge weakly to Y with $a = \Lambda(\{0\})$ and $\nu(dy) = y^{-2}\Lambda(dy)$ on $\mathbb{R}\setminus\{0\}$. Lower semicontinuity of $\int |y|\,\Lambda_n(dy) \le \beta_n - \sum_{k\le K} \mathbb{E}|X_{n,k}|^3$ as $n \to \infty$ followed by $K \to \infty$ yields $\beta \le \lim_n \beta_n$. □

Lemma 2.3 (Approximation of admissible representations by finite arrays)**.** *Every admissible representation S is a weak limit of finite independent centered unit-variance arrays whose third-moment sums converge to β. Consequently,*

$$\Delta(S) \le C_{\mathrm{ind}}\beta. \tag{2.2}$$

Proof. Truncate $(X_i)_{i\ge 1}$ to a finite prefix, approximate ν by a finite atomic measure $\sum_j \lambda_j \delta_{y_j}$, replace each Poisson atom $\lambda_j\delta_{y_j}$ by N independent variables $y_j(B - \lambda_j/N)$ ($B \sim \mathrm{Bern}(\lambda_j/N)$), and replace the remaining variance $b \to a$ by N independent symmetric two-point variables of variance b/N (whose third-moment sum is $b^{3/2}/\sqrt{N} \to 0$). Continuity of Φ implies $\Delta(S_N) \to \Delta(S)$, giving (2.2). □

Proposition 2.4 (Attainment of C_{ind} when $C_{\mathrm{ind}} > C_{\mathrm{E}}$)**.** *If $C_{\mathrm{ind}} > C_{\mathrm{E}}$, there exists a maximizing admissible representation whose discrete summands X_i are each supported on at most three points.*

Proof. Choose $c_n \uparrow C_{\mathrm{ind}}$ with $c_n > C_{\mathrm{E}}$, and apply Lemma 2.1 to obtain finite three-point unit-variance arrays with $\Delta(S_n) > c_n\beta_n$. By (1.1) and $\Delta(S_n) \le 1$, we have $\ell_0 < \beta_n < 1/c_n$. Passing to a subsequence with $\beta_n \to r \in [\ell_0, 1/C_{\mathrm{ind}}]$ via Lemma 2.2 gives a limiting representation S with $C_{\mathrm{ind}}r = \Delta(S) \le C_{\mathrm{ind}}\beta \le C_{\mathrm{ind}}r$, so $\beta = r > 0$ and $\Delta(S) = C_{\mathrm{ind}}\beta$. □

3 Variational identities, support size, and indecomposability

Fix a maximizing representation with $C = C_{\mathrm{ind}} > C_{\mathrm{E}}$ and threshold z satisfying (1.5). For any admissible perturbation of the uncentered summand laws with total mean m, central variance B, recentered third-moment functional R_*, and uncentered sum distribution function F_* at threshold w, Lemma 2.3 implies

$$J = B^{3/2}\left\{F_*(w) - \Phi\left(\frac{w-m}{\sqrt{B}}\right)\right\} - CR_* \le 0, \tag{3.1}$$

with $J = 0$ at the maximizing representation and $w = z$.

Lemma 3.1 (First-order contact inequalities)**.** *Let G_i be the distribution function of $S - X_i$, and put $v_i = \mathbb{E}X_i^2$, $\rho_i = \mathbb{E}|X_i|^3$, and $M_i = \mathbb{E}[X_i|X_i|]$. Then for all $u, y \in \mathbb{R}$,*

$$G_i(z-u) - F(z) + \phi(z)u + A(u^2 - v_i) - C\{|u|^3 - \rho_i - 3M_i u\} \le 0, \tag{3.2}$$

$$Q(y) := F(z-y) - F(z) + \phi(z)y + Ay^2 - C|y|^3 \le 0, \tag{3.3}$$

with equality in (3.2) at every $u \in \operatorname{supp} X_i$ and $Q(y) = 0$ at every $y \in \operatorname{supp}\nu \setminus \{0\}$. In particular, all supports $\operatorname{supp} X_i$ and $\operatorname{supp}\nu$ are uniformly bounded.

Proof. Perturbing $\mathrm{Law}(X_i)$ to $(1-t)\,\mathrm{Law}(X_i) + t\delta_u$ at $w = z - tu$ gives (3.2) (compare He and Cheng [7, Lemma 4.3]), and appending an independent compensated Poisson variable $y(N_t - t)$ at $w = z - ty$ gives (3.3). Since $-C|u|^3$ dominates as $|u| \to \infty$, all supports are uniformly bounded. □

For a mass transfer from v to u within X_i, write $d = u - v$, $b = u^2 - v^2$, and $g = G_i(z-u) - G_i(z-v)$. The first and second derivatives of $\mathbb{E}|X_i - \mathbb{E}X_i|^3$ are

$$\begin{aligned} r' &= |u|^3 - |v|^3 - 3d\mathbb{E}[X_i|X_i|], \\ r'' &= -6d(u|u| - v|v|) + 6d^2\mathbb{E}|X_i|. \end{aligned} \tag{3.4}$$

With $B_0 = [3D + (3z + z^3)\phi(z)]/4$ and $E_0 = (1 + z^2/2)\phi(z)$, the second derivatives of (3.1) with respect to mass transfers in coordinates i and j are

$$\begin{aligned} H_{ii} &= 3b_i g_i + B_0 b_i^2 + 2E_0 b_i d_i - 3Dd_i^2 - Cr_i'', \\ H_{ij} &= g_{ij} + \frac{3}{2}(b_i g_j + b_j g_i) + B_0 b_i b_j + E_0(b_i d_j + b_j d_i) + z\phi(z)d_i d_j \qquad (i \ne j), \end{aligned} \tag{3.5}$$

where g_{ij} is the mixed second difference of the raw distribution function.

3.1 Support size of every summand at a probability-local maximum

The next theorem applies unconditionally to any finite-array or admissible-representation maximizer (without requiring $C_{\mathrm{ind}} > C_{\mathrm{E}}$).

Theorem 3.2 (Four-point support bound and moment-preserving three-point replacement)**.** *Let (X_i) be the independent centered summands of a finite array or of an admissible representation with total variance $V > 0$ and third-moment functional $R > 0$. Suppose that at a threshold y with $D = \mathbb{P}(S \le y) - \Phi(y/\sqrt{V}) > 0$, the ratio $C = V^{3/2}D/R$ is locally maximal with respect to all two-sided bounded-density probability perturbations of each X_i that preserve its mass and mean. Then every summand X_i is supported on at most four points.*

Furthermore, if the array or representation is a global maximizer of the full Berry–Esseen ratio (at a fixed finite number of summands n, or over all admissible representations) and the selected branch attains the full Kolmogorov distance, then there exists an equally maximizing representation in which every summand is supported on at most three points and has the exact same variance $\mathrm{Var}(X_i)$ *and third absolute moment* $\mathbb{E}|X_i|^3$ *as before.*

Proof. Fix a summand X with law P, and let δ be a compactly supported signed measure with bounded density relative to P satisfying $\int d\delta = \int u\, d\delta = 0$. For small $|\varepsilon|$, $P + \varepsilon\delta$ is a centered probability measure with variance change $\varepsilon v_1 = \varepsilon \int u^2\, d\delta$, third-moment change $\varepsilon r_1 = \varepsilon \int |u|^3\, d\delta$, and affine raw CDF $F_\varepsilon(y) = \int \mathbb{P}(S - X \le y - u)\,(P + \varepsilon\delta)(du)$. Stationarity $C_0' = 0$ gives $D_0'/D = r_1/R - 3v_1/(2V)$, and with $z = y/\sqrt{V}$ and $q = z\phi(z)(3 - z^2)$, the second logarithmic derivative of $C_\varepsilon = (V + \varepsilon v_1)^{3/2} D_\varepsilon/(R + \varepsilon r_1)$ at $\varepsilon = 0$ is

$$\frac{C_0''}{C} = -\frac{15 + q/D}{4}\left(\frac{v_1}{V}\right)^2 + \frac{3 r_1 v_1}{RV} \le 0. \tag{3.6}$$

Consequently, the subspace of pairs (v_1, r_1) realizable by mean-zero, mass-zero perturbations δ cannot be all of $\mathbb{R}^2$. By Rolle's theorem applied twice to the second derivative $6|u| - 2c$, the four functions $1, u, u^2, |u|^3$ are linearly independent on any five distinct real points. If $\operatorname{supp} P$ contained five points, signed combinations of the restrictions of P to small disjoint neighborhoods of those points would realize every pair $(v_1, r_1) \in \mathbb{R}^2$, violating (3.6). Thus $|\operatorname{supp} P| \le 4$.

If P has four distinct atoms u_1, u_2, u_3, u_4 of positive masses p_j, then $(1, u, u^2, |u|^3)$ has rank 3 on $\{u_1, u_2, u_3, u_4\}$, so any nonzero vector (d_1, d_2, d_3, d_4) annihilating $1, u, u^2$ also annihilates $|u|^3$. Moving along $p_j + t d_j$ preserves mass, mean, variance, and $\mathbb{E}|X|^3$ while making $F(y)$ affine in t; stationarity at $t = 0$ forces $F(y)$ to be constant along the segment up to an endpoint where one atom mass vanishes, yielding a three-point summand with the exact same variance and third absolute moment. □

3.2 Convolution indecomposability of maximizing summands

By the $p = 3$ case of the inequality of Cox and Kemperman [4, Theorem (2.6)] (see also Ruzankin [13, Theorem E and Sections 2.1–2.2]), third absolute moments are strictly superadditive under convolution of nonconstant independent centered random variables:

Lemma 3.3 (Strict third-moment superadditivity)**.** *For any independent centered real random variables U, W with finite third absolute moments,*

$$\mathbb{E}|U + W|^3 \ge \mathbb{E}|U|^3 + \mathbb{E}|W|^3,$$

with equality if and only if at least one of U, W is identically zero.

Proof. For centered two-point variables on $\{-a, b\}$ and $\{-c, d\}$ $(a, b, c, d > 0)$, multiplying $\mathbb{E}|U + W|^3 - \mathbb{E}|U|^3 - \mathbb{E}|W|^3$ by $(a+b)(c+d)$ gives a function $N(a, b, c, d)$ that vanishes when any argument is 0 and satisfies $\partial_a\partial_b\partial_c\partial_d N = 12\{\min(a, d) + \min(b, c)\} > 0$ on $(0, \infty)^4$, so $N > 0$. Decomposing any nonconstant centered distribution into a mixture of δ_0 and centered two-point laws via $M(da, db) = (a + b)\mu_-(da)\mu_+(db)/\mathbb{E}U_+$ proves strict inequality in general. □

Corollary 3.4 (Convolution indecomposability)**.** *Every nonconstant summand X_i of a global maximizer over admissible representations (or of a fixed-n global maximizer having at least one zero summand) is indecomposable as a convolution $U + W$ of two nonconstant independent real random variables. In particular, no summand can be a three-point distribution supported on an arithmetic progression $\{0, g, 2g\}$ $(g > 0)$ with atom probabilities (a, b, c) satisfying $b^2 \ge 4ac$.*

Proof. Replacing a decomposable summand $X_i \overset{d}{=} U + W$ by the two independent centered summands U, W leaves the sum S and total variance unchanged while strictly reducing the third-moment sum R by Lemma 3.3. A three-point law on $\{0, g, 2g\}$ with weights (a, b, c) factors as the convolution of two Bernoulli laws on $\{0, g\}$ if and only if $t^2 - (b + 2c)t + c = 0$ has real roots in $(0, 1)$, which is equivalent to $b^2 \geq 4ac$. □

3.3 Point masses of infinite sums via Sperner's theorem

Lemma 3.5. *If $S = \sum_{i \geq 1} X_i + Y$ has an atom of mass $m > 0$ and each X_i is supported on at most three points, then with $q_i = \max_{u \in \mathbb{R}} \mathbb{P}(X_i = u)$,*

$$\sum_{i \geq 1} (1 - q_i) < \infty.$$

Proof. Each three-point law X_i can be written as a mixture in which, with probability $\rho_i = \min\{1, 2(1 - q_i)\} \geq \frac{3}{2}(1 - q_i)$, a fair symmetric two-point step is activated. Conditional on the number N of activated fair steps and their step sizes, every level set of the sum is an antichain in $\{0, 1\}^N$, so Sperner's theorem [15] bounds its conditional probability by $2^{-N}\binom{N}{\lfloor N/2 \rfloor} \leq \sqrt{2/(N+1)}$; averaging over N gives $m^2 \leq 2\mathbb{E}[(N+1)^{-1}] \leq 2(1 - e^{-\mu})/\mu$ where $\mu = \sum_i \rho_i$ (compare Jessen and Wintner [9]). □

4 Eliminating Lévy jumps and nonshrinking supports

4.1 Pair contacts and finiteness of the Lévy support

Lemma 4.1. *For any two same-sign points $x, y \in \operatorname{supp} \nu \setminus \{0\}$, one has $Q(x + y) = 0$. Moreover, $\operatorname{supp} \nu \setminus \{0\}$ contains at most two positive and two negative points.*

Proof. Replacing a compensated Poisson component of intensity $p - kp^2$ ($k = m^2/v$) and jump law H supported on one side of 0 by a single centered variable $B_p H - pm$ preserves total variance to second order in p and expands the objective excess at $O(p^2)$ as $(k+1)\mathbb{E}Q(H) - \frac{1}{2}\mathbb{E}Q(H + H') \leq 0$; since $Q = 0$ on $\operatorname{supp} \nu \setminus \{0\}$, upper semicontinuity forces $Q(x+y) = 0$ for all same-sign $x, y \in \operatorname{supp} \nu \setminus \{0\}$.

Perturbing ν by $\eta = f\nu$ on one side $\sigma \in \{-1, 1\}$ of 0 gives

$$(1 + v_\eta)^{3/2} \left\{ [e^{U_\eta} F](z) - \Phi\left(\frac{z - d_\eta}{\sqrt{1 + v_\eta}} \right) \right\} - C(\beta + r_\eta) \leq 0. \tag{4.1}$$

At same-sign contacts $x = \sigma u, y = \sigma v$ whose sum $x + y$ is also a contact of Q, the second-order intensity kernel is

$$\mathcal{K}_\sigma(u, v) = uv \left\{ -3D + \left(3C + \frac{\sigma \phi''(z)}{2} \right)(u + v) - Luv + \frac{3C}{2} uv(u + v) \right\}, \qquad u, v > 0, \tag{4.2}$$

where $L = (15D + \phi'''(z))/4$. In the moment basis (u, u^2, u^3), the coefficient matrix of $\mathcal{K}_\sigma$ is

$$M_\sigma = \begin{pmatrix} -3D & 3C + \sigma\phi''(z)/2 & 0 \\ 3C + \sigma\phi''(z)/2 & -L & 3C/2 \\ 0 & 3C/2 & 0 \end{pmatrix}, \tag{4.3}$$

which is not negative semidefinite since $(0, t, 1) M_\sigma (0, t, 1)^\mathsf{T} = -Lt^2 + 3Ct > 0$ for small $t > 0$. By Vandermonde invertibility of $(u_j, u_j^2, u_j^3)_{j=1}^3$, ν cannot have three distinct support points of the same sign. □

4.2 Contact propagation and proof of Proposition 4.4

Lemma 4.2. *If i_n are distinct indices and $u_n \in \operatorname{supp} X_{i_n}$ satisfies $u_n \to x \neq 0$, then $Q(x) = Q(2x) = 0$. Moreover, the mass transfer can be taken from a state $v_n \to 0$ to u_n with gap $d_n = u_n - v_n \to x$, and if F has atoms and $d_n > x$ along a subsequence, then F is continuous at $z - x$ and $z - 2x$.*

Proof. Since $v_{i_n} \to 0$, choosing a modal state $v_n \to 0$ (via Lemma 3.5 if F has atoms) gives $Q(x) = 0$ from first-order stationarity and $-2Q(2x) \leq 0$ from the two-coordinate Hessian contrast $(1, -1)$ in (3.5), which together with $Q(2x) \leq 0$ forces $Q(2x) = 0$. □

Lemma 4.3. *Let $x = \sigma u \neq 0$ be either a limit of support points $u_n \in \operatorname{supp} X_{i_n}$ or an atom of ν, and let $y = \sigma v \neq 0$ be any fixed same-sign contact point with $Q(y) = 0$. Then the limiting mixed Hessian entry between the old two-sided direction at x and an adjoined summand $(1 - t)\delta_0 + t\delta_y$ $(t \geq 0)$ equals $\mathcal{K}_\sigma(u, v) + Q(x + y)$, while both the old diagonal entry and its mixed entry with an adjoined summand at x equal $\mathcal{K}_\sigma(u, u)$.*

Proof. Comparing the old direction at x, an adjoined summand at x, and an adjoined summand at y along the tangent direction $(-1, 1, 0) + t(0, 0, 1)$ $(t \geq 0)$, where $(-1, 1, 0)$ has zero quadratic value, forces the old–adjoined-y mixed entry to be at least—and by upper semicontinuity equal to—the closed adjoined-x–adjoined-y mixed entry $\mathcal{K}_\sigma(u, v) + Q(x + y)$. □

Proposition 4.4. *Every maximizing representation with three-point discrete summands satisfies $\nu = 0$ and $\max\{|u| : u \in \operatorname{supp} X_i\} \to 0$ as $i \to \infty$.*

Proof. Suppose $x = \sigma u \neq 0$ is either an atom of ν or a limit of $u_n \in \operatorname{supp} X_{i_n}$, and let $y = \sigma v$ be any same-sign contact with $Q(y) = 0$. For fixed $\varepsilon > 0$, adjoin a single three-point summand $(1 - (1 + \varepsilon)t)\delta_0 + t\delta_x + \varepsilon t\delta_y$ $(t \geq 0)$, which has first derivative $Q(x) + \varepsilon Q(y) = 0$. Because both x and y lie in the same adjoined summand, its raw CDF is affine in t (with no double-shift term on its diagonal), so its diagonal Hessian entry is $\mathcal{K}_\sigma(u, u) + 2\varepsilon\mathcal{K}_\sigma(u, v) + \varepsilon^2\mathcal{K}_\sigma(v, v)$, while by Lemma 4.3 its mixed entry with the old two-sided direction at x is $\mathcal{K}_\sigma(u, u) + \varepsilon[\mathcal{K}_\sigma(u, v) + Q(x + y)]$. Testing the 2×2 Hessian on $(-1, 1)$ gives $-2\varepsilon Q(x + y) + \varepsilon^2\mathcal{K}_\sigma(v, v) \leq 0$; sending $\varepsilon \downarrow 0$ forces $Q(x + y) \geq 0$ and hence $Q(x + y) = 0$. By induction on $n \geq 1$ with $y = nx$, we obtain $Q(nx) = 0$ for all $n \geq 1$, contradicting the boundedness of the zero set of Q in Lemma 3.1. □

5 Eliminating the Gaussian component

By Proposition 4.4, $S = W + G_a$ where $W = \sum_i X_i$ has variance $1 - a$, $\beta = \sum_i \mathbb{E}|X_i|^3$, and $G_a \sim N(0, a)$ is independent. If $a > 0$, the smooth density f of S satisfies threshold and Gaussian-variance stationarity:

$$f(z) = \phi(z), \qquad f'(z) = -3D - z\phi(z), \qquad |f''(z)| \leq 6C. \tag{5.1}$$

5.1 Zero-bias Ornstein–Uhlenbeck bound and lattice comparison

By (5.2) and the Goldstein–Tyurin bound

$$W_1(W, W^*) \leq \frac{\beta}{2(1 - a)}, \tag{5.2}$$

solving the Stein equation for $h(u) = \Phi((z - u)/\sqrt{a})$ gives:

Lemma 5.1. *For every $0 < a < 1$,*

$$\Delta(W + G_a) \le K(a)\beta, \qquad K(a) = \frac{\phi(0)}{2} \int_0^1 \frac{s^2}{[1-(1-a)s^2]^{3/2}}\, ds.$$

Since $K(a)$ is decreasing and $K(1/10) < C_{\mathrm{E}}$, any maximizing representation satisfies $a < 1/10$.

Lemma 5.2. *At a maximizing representation with $a > 0$,*

$$\phi(z) \le a\left(C - \frac{f''(z)^2}{36C}\right). \tag{5.3}$$

Proof. Replacing the tail $\sum_{i>K} X_i$ by its accompanying Poisson law and rounding the summands and jumps to a fine lattice $h_j\mathbb{Z}$ produces admissible approximations $T_j + G_{a_j}$ ($a_j \to a$) with $T_j \in \theta_j + h_j\mathbb{Z}$ and

$$\beta_j = \beta + o(h_j), \qquad \|F_{T_j+G_{a_j}} - F\|_\infty = o(h_j), \qquad f_j^{(k)} \to f^{(k)} \text{ uniformly} \quad (k = 0, 1, 2). \tag{5.4}$$

Replacing G_{a_j} by a centered binomial $B_j = h_j(\mathrm{Bin}(n_j, p_j) - n_j p_j)$ of variance a_j with $p_j \to p \in (0, 1)$ ($u_j = 1 - 2p_j$) and applying Esseen's lattice Edgeworth expansion [5, eq. (3)]

$$\mathbb{P}(B_j \le x) = \Phi_{a_j}(x) + \frac{h_j}{2}\phi_{a_j}(x) - \frac{a_j h_j u_j}{6}\phi''_{a_j}(x) + o(h_j) \tag{5.5}$$

at a lattice point z_j within $h_j/2$ of z yields $\phi(z) - \frac{1}{3}auf''(z) - Ca(1 + u^2) \le 0$ for all $u = 1 - 2p \in [-1, 1]$. Optimizing at $u = -f''(z)/(6C)$ proves (5.3). □

Combining (5.3) with Gaussian convolution semiconvexity $f'(z)^2 \le f(z)f''(z) + f(z)^2/a$ gives $\beta < \frac{17}{20}\sqrt{a}$, which together with Lemma 5.1 confines (a, β) to the window

$$0 < \beta < \frac{269}{1000}, \qquad \frac{400}{289}\beta^2 < a < \frac{1}{10}. \tag{5.6}$$

5.2 Fourier exclusion of the remaining Gaussian window

By Korolev and Shevtsova [10, Section 3.3, Theorem 2], any centered infinitely divisible law Y with variance $v > 0$ and third absolute jump integral r satisfies

$$d_{\mathrm{K}}(Y, N(0, v)) \le 0.3051\, \frac{r}{v^{3/2}}. \tag{5.7}$$

Lemma 5.3. *Every unit-variance representation $S = W + G_a$ satisfying (5.6) obeys $\Delta(S) < (2/5)\beta < C_{\mathrm{E}}\beta$.*

Proof. Using the global inequality

$$\cos x \le 1 - \frac{x^2}{2} + \frac{|x|^3}{10} \qquad (x \in \mathbb{R}) \tag{5.8}$$

and $\mathbb{E}|X_i - X_i'|^3 \le 4\mathbb{E}|X_i|^3$, each summand characteristic function $h_i(t)$ satisfies

$$|h_i(t)| \le \exp\left\{-\frac{v_i t^2}{2} + \frac{b_i|t|^3}{5}\right\}, \qquad b_i = \mathbb{E}|X_i|^3, \tag{5.9}$$

and integrating the zero-bias differential inequality gives

$$|h(t) - e^{-t^2/2}| \le \frac{\beta}{6} e^{25/54} t^3 e^{-t^2/2+\beta t^3/5} \qquad (t \ge 0). \tag{5.10}$$

Splitting the exact Fourier inversion integral for $S = W + G_a$ at $T = c/\beta > 7$ ($c = 19/10$) bounds the low-frequency part $[0, T]$ by

$$\frac{1}{\pi\beta} \int_0^T \frac{|h(t) - e^{-t^2/2}|}{t}\, dt \le \frac{e^{25/54}}{6\pi}(I + J), \tag{5.11}$$

with $I = \int_0^6 t^2 e^{-t^2/2+(269/5000)t^3}\, dt < 23/10$ and $J = \int_6^\infty t^2 e^{-3t^2/25}\, dt < 3/8$, while on $[T, \infty)$ domination of $|h_W(t)|$ by its symmetrized accompanying compound-Poisson characteristic function and (5.7) bound the tail by $(11/19)e^{-361/289} + 10^{-9} < 1/6 + 10^{-9}$. The rational checks in Appendix A give $\Delta(S)/\beta < 1182125003/3000000000 < 2/5$, proving $a = 0$. □

6 Nonatomicity of infinitely active maximizing sums

Suppose $S = \sum_{i \ge 1} X_i$ has an atom, and let c_i be a modal state of X_i with probability $q_i = 1 - p_i \ge 1/3$, so $\sum_i p_i < \infty$ by Lemma 3.5 and $\sum_i |c_i| < \infty$. If infinitely many X_i were nonconstant, choose K with $q_j \ge 3/4$ for $j \ge K$, and for a late nonconstant coordinate $i \ge K$ with nonmodal state $u_i = c_i + d_i$ and $S_i^0 = S - X_i + c_i$, its mass-transfer derivative is

$$g_i = -\operatorname{sgn}(d_i)\mathbb{P}(E_i), \qquad E_i = \begin{cases} \{z - d_i < S_i^0 \le z\}, & d_i > 0, \\ \{z < S_i^0 \le z - d_i\}, & d_i < 0. \end{cases} \tag{6.1}$$

Letting N_i count the nonmodal outcomes among $j \ge K$, $j \ne i$, compactness of finite configurations implies that for every fixed $M \ge 1$,

$$E_i \subset \{N_i > M\} \qquad \text{for all sufficiently large } i. \tag{6.2}$$

Meanwhile, stationarity of the mass transfer gives

$$\frac{g_i}{d_i} \longrightarrow -\phi(z), \qquad \frac{\mathbb{P}(E_i)}{|d_i|} \longrightarrow \phi(z) > 0. \tag{6.3}$$

Pairing the mass transfer s in coordinate i with the collective product-law variation t that multiplies all nonmodal probabilities for $j \ge K, j \ne i$ by $1 + t$, the 2×2 Hessian of (3.1) satisfies

$$|J_{ss}| \le K_1 d_i^2, \qquad |J_{st} - g_{i,t}| \le K_2 |d_i|, \qquad |J_{tt}| \le K_3, \tag{6.4}$$

whereas $|g_{i,t}|/|d_i| = \mathbb{E}[L_i \mid E_i]\mathbb{P}(E_i)/|d_i| \to \infty$ by (6.2)–(6.3), contradicting $|J_{st}|^2 \le J_{ss} J_{tt} \le K_1 K_3 d_i^2$. This completes the proof of Theorem 1.2.

7 Support motions and completion rank

To prove Theorem 1.5, suppose that for arbitrarily late finite sets I there exist nonconstant centered functions $f_i(X_i)$ ($\mathbb{E} f_i(X_i) = 0$, $w_i = \mathbb{E} f_i(X_i)^2$, $W = \sum_{i \in I} w_i > 0$, $V = \sum_{i \in I} v_i \to 0$, and $h = \max_{i \in I} \max |\operatorname{supp} X_i| \to 0$) with $\sum_{i \in I} f_i(x_i) = c$ on $\mathcal{A}_I$. Moving each support state x_i to $x_i + t f_i(x_i)$

and the threshold to $z+tc$ leaves the raw event probability $F_*(z+tc)$ identically constant for small $|t|$. Choosing $j \in I$ to maximize w_j/v_j, so that

$$\frac{v_j}{w_j} \le \frac{V}{W}, \tag{7.1}$$

and tilting $\mathrm{Law}(X_j)$ by $(1+sf_j(X_j))$ while applying the support motion t, the centered threshold x, variance B, and third-moment functional R have derivatives at $(s,t)=(0,0)$ given by

$$\begin{aligned} x_s = -m, \quad x_t = c, \quad x_{st} = -w, \qquad B_s = a_1, \quad B_{ss} = -2m^2, \\ B_t = 2M, \quad B_{tt} = 2W, \qquad B_{st} = 2b_1, \end{aligned} \tag{7.2}$$

and

$$\begin{aligned} R_t &= 3\sum_{i\in I} \mathbb{E}[X_i|X_i|f_i], & R_{tt} &= 6\sum_{i\in I} \mathbb{E}[|X_i|f_i^2], \\ R_s &= \mathbb{E}[|X|^3 f] - 3m\mathbb{E}[X|X|], & R_{ss} &= 6m^2\mathbb{E}|X| - 6m\mathbb{E}[X|X|f], \\ R_{st} &= 3\mathbb{E}[X|X|f^2] - 3w\mathbb{E}[X|X|] - 6m\mathbb{E}[|X|f]. \end{aligned} \tag{7.3}$$

Using (7.1)–(7.3), the 2×2 Hessian of (3.1) satisfies $|J_{ss}| \le K_1 vw$, $|J_{tt}| \le K_2 W$, and $J_{st} = \phi(z)w + O((h+V)w)$, which violates $J_{st}^2 \le J_{ss}J_{tt} \le K_1K_2Vw^2$ as $h+V \to 0$. This proves Theorem 1.5. Corollary 1.6 follows immediately by noting that if (1.7) held, at most $2N$ block-sum vectors among $M > 2N$ late blocks could belong to $\mathcal{A}_I$, yielding a nontrivial linear relation $\sum_{k=1}^{M} a_k u_k = c$ on $\mathcal{A}_I$.

8 Support geometry and weighted variance setup

We now assume the third-moment upper bound

$$\beta \le \beta_* := \frac{18481}{25000} = 0.73924 \tag{8.1}$$

and prove Theorem 1.3 and Corollary 1.4. Throughout Sections 8–12, we consider a maximizing representation with at most three support points per summand as in Theorem 1.2 and threshold data (1.5). Omitting deterministic zero summands, write

$$H_i = \mathrm{diam}(\mathrm{supp}\, X_i), \qquad v_i = \mathbb{E}X_i^2, \qquad V(h) = \sum_{H_i \le h} v_i, \qquad W_s = \sum_{i\ge 1} v_i H_i^{-s}.$$

Lemma 8.1 (Strict scalar curvature margin)**.** *Let $C \ge C_{\mathrm{E}}$, $D = C\beta$, and define*

$$L_0 = \frac{15D + \phi'''(z)}{4}, \qquad c_\varepsilon = 3C + \frac{\varepsilon\phi''(z)}{2} \qquad (\varepsilon \in \{-1,1\}).$$

Under (8.1), $c_\varepsilon > 0$ and $c_\varepsilon^2 > 3DL_0$ for both $\varepsilon \in \{-1,1\}$.

Proof. With $r = 486833/500000 > \phi(0)/C_{\mathrm{E}} = 6/(3+\sqrt{10})$ and $x = |z|$, $(c_\varepsilon^2 - 3DL_0)/C^2$ is bounded below by

$$\left[3 - \frac{r}{2}|x^2-1|e^{-x^2/2}\right]^2 - \frac{3\beta_* r}{4}|3x - x^3|e^{-x^2/2} - \frac{45}{4}\beta_*^2.$$

On $0 \le x \le 1/2$, replacing $e^{-x^2/2}$ by $T(x) = 1 - x^2/2 + x^4/8 - x^6/48 + x^8/384$ gives a degree-20 polynomial $P(x)$ whose Bernstein coefficients on the five dyadic subintervals $[0,1/8]$, $[1/8,3/16]$, $[3/16,7/32]$, $[7/32,1/4]$, and $[1/4,1/2]$ exceed $1/40$, $1/1000$, $1/30000$, $1/1000$, and $9/1000$, respectively. For $x \ge 1/2$, $|\phi''(x)|/\phi(0) < 2/3$ and $|\phi'''(x)|/\phi(0) < 7/5$ give the lower bound $(3-r/3)^2 - \frac{21}{20}\beta_* r - \frac{45}{4}\beta_*^2 = 35773617637/140625000000 > 0$. □

Lemma 8.2 (Balanced extreme support points for three-point summands)**.** *There exists $\kappa > 0$ such that every sufficiently late nondegenerate three-point summand X_i, with extreme support points $-a_i, b_i$ $(a_i, b_i > 0)$, satisfies* $\min(a_i, b_i) \geq \kappa H_i$*. Consequently, for all $i \geq 1$,*

$$H_i \mathbb{E}|X_i| \leq K v_i. \tag{8.2}$$

Proof. For a three-point summand with support $x < y < t$ in $[-h, h]$, varying its raw mean and second moment on the fixed support gives the second variation

$$(-3D - Cr_{\mu\mu})e^2 + (\phi''(z) + 6Ck + 3Cr_\mu)ef + (-L_0 + 3Cr_v)f^2 \leq 0,$$

where $k = [u|u|]_{x,y,t} \in [-1, 1]$ is the second divided difference of $u|u|$ and $r_\mu, r_v, r_{\mu\mu} = O(h)$. As $h \to 0$, nonpositivity of this quadratic form requires $(3Ck + \phi''(z)/2)^2 \leq 3DL_0$, which by Lemma 8.1 forces $|k|$ to stay bounded away from 1. Since $k = 1 - 2a^2/[(a+y)(a+b)]$ for $y \geq 0$ and $k = -1 + 2b^2/[(a+b)(b-y)]$ for $y \leq 0$, bounding $|k|$ away from 1 forces $\min(a, b) \geq \kappa H$, which immediately implies $v_i \geq \frac{1}{2}\min(a_i, b_i)\mathbb{E}|X_i| \geq \frac{1}{2}\kappa H_i \mathbb{E}|X_i|$. □

9 Third-moment growth under grouping and normal comparison

When disjoint summands are grouped into a partial sum, the sum law S and total variance are unchanged, while the sum of third absolute moments increases by the *grouping cost*.

Lemma 9.1 (Variance and annular grouping bounds)**.** *For any independent centered family (X_i) with total variance V,*

$$0 \leq \mathbb{E}\left|\sum_i X_i\right|^3 - \sum_i \mathbb{E}|X_i|^3 \leq 2V^{3/2}. \tag{9.1}$$

More precisely, if $X_i = Y_i - \mathbb{E}Y_i$ with $M = \sum_i \mathbb{E}|Y_i|$ and $A_- = \sum_i \mathbb{E}(Y_i)_-$, then the grouping cost is at most $2M^3 + 6A_-V$. Consequently, if the summands are partitioned into B classes with first-moment and variance sums M_j, V_j, then for every integer $J \geq 1$ there exists a partition into at most $8J + 6B$ groups whose total grouping cost is at most

$$\frac{18M^3}{J^2} + \frac{9S^2}{J}, \qquad M = \sum_{j=1}^{B} M_j, \qquad S = \sum_{j=1}^{B} \sqrt{M_j V_j}. \tag{9.2}$$

Proof. Inequality (9.1) follows by summing the incremental bound $\mathbb{E}|S_{<i} + X_i|^3 - \mathbb{E}|S_{<i}|^3 - \mathbb{E}|X_i|^3 \leq 3v_i\sqrt{V_{<i}}$ (compare Pinelis [12, eq. (5)]). The refined bound $2M^3 + 6A_-V$ follows from the exact identity $\mathbb{E}|X+Y|^3 - \mathbb{E}|X|^3 - \mathbb{E}|Y|^3 = 12\int_{\mathbb{R}} k_X(t)k_Y(-t)\,dt$, where $k_X(t) = \mathbb{E}(X-t)_+ - (-t)_+ \geq 0$ has $\int_{\mathbb{R}} k_X(t)\,dt = v_X/2$, $\sup_{t\in\mathbb{R}} k_{X_i}(t) \leq M_i$, and $k_{X_i}(-t) \leq (M_i - t)_+ + A_i$ for $t \geq 0$. Greedy subdivision within each class j with budget $J_j = \lceil J(M_j/M + \sqrt{M_jV_j}/S)\rceil$ yields (9.2). □

Lemma 9.2 (Quadratic majorant for a grouped block)**.** *Let U be the sum of a subcollection of summands of variance $\sigma^2 = \mathbb{E}U^2 > 0$ and maximum summand radius H, independent of the remainder with distribution function G. There exists a quadratic polynomial $P(u) = a + bu + cu^2$ such that*

$$Q_U(u) := P(u) + C|u|^3 - G(z-u) \geq 0 \quad (u \in \mathbb{R}), \qquad \mathbb{E}Q_U(U) \leq 2C\sigma^3. \tag{9.3}$$

As $\sigma \to 0$ with $H/\sigma \to 0$, its coefficients satisfy

$$|a| + |b| + |c| \leq K, \qquad b + \phi(z) = O(\sigma), \qquad c + A = O(\sigma). \tag{9.4}$$

Proof. By strong moment duality at $(0,\sigma^2)$, the supporting plane $P(u)$ to the concave value function of $\sup\{\mathbb{E}[G(z-Y)-C|Y|^3] : \mathbb{E}Y = m, \mathbb{E}Y^2 = s\}$ majorizes $G(z-u)-C|u|^3$ and satisfies $\mathbb{E}Q_U(U) \le C(\mathbb{E}|U|^3 - \beta_I) \le 2C\sigma^3$ by (9.1). Testing two-sided bounded-density perturbations $(1+t\psi)\,\mathrm{Law}(U)$ orthogonal to $1, U$ gives

$$\mathbb{E}|G(z-U) - F(z) + \phi(z)U| \le K\sigma^2 \tag{9.5}$$

and $\mathbb{E}|G(z-U) - C|U|^3 - a_0 - b_0U + AU^2| = O(\sigma^3)$ by L^1 quotient duality, from which (9.4) follows. □

Lemma 9.3 (Weighted normal comparison via Cramér moderate deviations)**.** *In Lemma 9.2, if* $(H/\sigma)\log^{3/2}(1/\sigma) \to 0$*, then for* $Z \sim N(0,1)$*,*

$$\mathbb{E}Q_U(\sigma Z) \le K(\sigma^3 + H). \tag{9.6}$$

If $H/\sigma = O(\sigma^\zeta)$ *for some fixed* $\zeta > 1$*, then for every* $2 < q < \min(\zeta+1, 3)$ *and fixed* $T > 0$*,*

$$\sup_{|t|\le T\sigma} \left|F(z+t) - F(z) - \phi(z)t + At^2\right| = O_{q,T}(\sigma^q). \tag{9.7}$$

Proof. For a variance-one sum $W = U/\sigma$ of independent centered variables bounded by $\epsilon = H/\sigma$, Cramér exponential tilting (see also Beknazaryan et al. [1, Theorem 2.1]) gives

$$|\mathbb{P}(W \le x) - \Phi(x)| + |\mathbb{P}(W < x) - \Phi(x)| \le K\epsilon(1+x^2)\phi(x) \qquad (|x| \le L). \tag{9.8}$$

Integrating $Q_\sigma(x) = Q_U(\sigma x) \ge 0$ by parts against $(1+x^2)\phi(x)$ on $[-L, L]$ ($L = \sqrt{20\log(1/\sigma)}$) via (9.8) proves (9.6), and applying the same bound to $Q_\sigma(x-t)$ gives

$$\sup_{|t|\le T} \mathbb{E}Q_U(U - \sigma t) \le \sigma^{-o(1)}(\sigma^3 + H). \tag{9.9}$$

Multiplying the individual contact equations by X_i and summing refines $b + \phi(z) = O(\sigma^2\sqrt{\log(1/\sigma)})$, which together with (9.9) proves (9.7). □

9.1 Randomized lattice phase identity and small-diameter Fourier bound

For $N \ge 3$ independent groups rounded unbiasedly to a lattice of mesh ℓ with uniform phases $u_1, \dots, u_{N-1}$ and $u_N \equiv z - \sum_{i<N} u_i \pmod{\ell}$, the phase-averaged rounded distribution function at z satisfies the exact identity

$$\mathbb{E}_{\boldsymbol{u}} F_{L_{\boldsymbol{u}}}(z) = \mathbb{E}F(z - W_\ell), \qquad W_\ell = T_1 + \cdots + T_{N-1} - U_\ell, \tag{9.10}$$

where T_i have independent triangular densities $\ell^{-1}(1-|x|/\ell)_+$ and $U_\ell \sim \mathrm{Unif}[0,\ell]$, so $\mathbb{E}W_\ell = -\ell/2$ and $\mathbb{E}W_\ell^2 = (N+1)\ell^2/6$, while the total rounding variance $v_{\boldsymbol{u}}$ satisfies

$$\mathbb{E}v_{\boldsymbol{u}} = \frac{N\ell^2}{6}, \qquad \mathrm{Var}(v_{\boldsymbol{u}}) \le \frac{N\ell^4}{64}, \qquad |\,\mathrm{Cov}(v_{\boldsymbol{u}}, F_{L_{\boldsymbol{u}}}(z))| \le \frac{\sqrt{N}\ell^2}{16} \tag{9.11}$$

(using Hoeffding's inequality [8] to control tails of W_ℓ). Consequently, we obtain:

Lemma 9.4 (Phase-rounding exclusion at a normal window)**.** *Suppose* (9.7) *holds with* $2 < q < 3$ *along* $\sigma \downarrow 0$*, and the summands are partitioned into* $N \to \infty$ *groups with grouping cost* $w \ge 0$*. Then maximality is violated whenever, with* $L = \log(1/\sigma)$*,*

$$N\sigma^{2q-2}L \longrightarrow 0, \qquad \frac{w\sqrt{NL}}{\sigma} \longrightarrow 0, \qquad wN^{2/3} \longrightarrow 0. \tag{9.12}$$

Lemma 9.5 (Fourier exclusion of slow small-diameter variance decay)**.** *No maximizing representation can satisfy $V(r) \ge cr^p$ for all sufficiently small $r > 0$ with an exponent $p < 2$.*

Proof. If $V(r) \ge cr^p$ with $p < 2$, then $w(t) = \exp\{-t^2 V(1/(8t))/12\}$ has finite moments of all orders on $[1, \infty)$. Rounding the first k summands and interpolating the tail's accompanying Poisson second-moment measure to $\ell_k\mathbb{Z}$ ($\ell_k = \sqrt{\sum_{i>k} v_i^2/k}$) yields lattice approximations Z_k with $F_{Z_k}(x) = F(x) + \frac{1}{2}\ell_k\phi(z) + o(\ell_k)$ at a lattice point within $\ell_k/2$ of z and variance and third-moment changes $o(\ell_k)$, producing a positive objective gain $\frac{1}{2}\phi(z)\ell_k + o(\ell_k) > 0$. □

10 Initial weighted variance moments ($W_{1/2}$ and $W_{3/4}$)

Lemma 10.1. *Under (8.2), both $W_{1/2}$ and $W_{3/4}$ are finite.*

Proof. By Tonelli's theorem,

$$\sum_{H_i \le 1} v_i H_i^{-s} = V(1) + s\int_0^1 V(u)u^{-s-1}\,du. \tag{10.1}$$

If $W_{1/2} = \infty$, then points with $V(u) \ge u^a$ ($a = 13/24$) occur arbitrarily close to 0, while Lemma 9.5 supplies $r \downarrow 0$ with $V(r) \le r^p$ ($p = 15/8$). Taking the last crossing $h < r$ where $V(h) = h^a$ and $V(u) < u^a$ on $(h, r]$ and applying Lemma 9.4 at $\sigma \sim h^{1/3}$, $q = 23/8$, and $t = h^{37/72}$ yields a contradiction. Once $W_{1/2} < \infty$, the annular grouping criterion

$$M\sigma^{q-4/3}L^{2/3} \to 0, \qquad S^2\sigma^{q-2}L \to 0, \qquad \frac{V^3 L}{\sigma^2} \to 0, \qquad B\sigma^{2q-2}L \to 0 \tag{10.2}$$

at $a = 7/8$, $p = 15/8$, $Q_0 = 15/7$, $q = 29/14$, and $t = \sqrt{h}$ similarly rules out $W_{3/4} = \infty$. □

11 Compound-Poisson intensity bounds on dyadic shells

Set $d(u) = -Q(u) = F(z) - \phi(z)u - Au^2 + C|u|^3 - F(z-u) \ge 0$, and for $m > 3$ define the monotone envelope

$$E_m(h) = h^3 + \sup_{r \ge h}\left(\frac{h}{r}\right)^m \sup_{|u| \le r} d(u). \tag{11.1}$$

Lemma 11.1 (Pairwise contact defect bound)**.** *Let two distinct summands with modal states translated to 0 have laws $K_i = (1-\theta_i)\delta_0 + \eta_i$ ($\theta_i \le 1/4$), diameters at most Lh, and same-sign states $x = \varepsilon u$, $y = \varepsilon v$ with $\kappa h \le u, v \le Lh$. Under Lemma 8.1,*

$$d(x) \le K\theta_1 E_m(h), \qquad d(y) \le K\theta_2 E_m(h), \qquad d(x+y) \le K(\theta_1+\theta_2)E_m(h).$$

Proof. Inverting K_i via its Neumann series and testing the 2×2 probability Hessian on $(1+t, -1)$ with $t = c_\varepsilon(u-v)/(3D)$ gives $uv(u-v)^2 \le K(\theta_1+\theta_2)E_m(h)$ by Lemma 8.1; testing on $(1, -1)$ then bounds $d(x+y)$. □

Lemma 11.2 (Shell nonmodal intensity bound)**.** *Fix $\kappa, L > 0$. For finitely many summands with modal state 0, diameters at most Lh, and modal deficits $\theta_i \le \epsilon$, let submeasures ν_i of their nonmodal states in $\varepsilon[\kappa h, Lh]$ have total mass $\lambda = \sum_i \nu_i(\mathbb{R}) \in (0, 1]$. Then for small h, ϵ,*

$$\epsilon E_m(h) \ge c\lambda h^4 \min(\lambda, h), \qquad \lambda \le K\{\sqrt{\epsilon E_m(h)}\, h^{-2} + \epsilon E_m(h) h^{-5}\}. \tag{11.2}$$

Proof. With $\mu = \lambda^{-1}\sum_i \nu_i$, $a = \int x\, d\mu \ge \kappa h$, $s = 2c_\varepsilon a/(3D)$, and $\alpha = 2 - s$, adjoin one compound-Poisson-type summand $\varepsilon(\sum_{j=1}^{N_q} Z_j - 2qa)$ whose count N_q has probability generating function

$$B_q(t) = e^{\alpha q(t-1)}\big[1 + sq(t-1) + q(t-1)^2\big], \tag{11.3}$$

while withdrawing fraction $\alpha q/\lambda$ of each ν_i to its mode 0. In the weighted convolution norm $\|\xi\|_m = \int (1+|u|/h)^m\, |\xi|(du)$, the relative convolution operator $\mathcal{P}(q) = \prod_i (I - \alpha q A_i) e^{\alpha q D_\mu}[I + q(sD_\mu + D_\mu^2)]$ satisfies

$$\|\mathcal{P}(q) - I - q\mathcal{P}'(0)\|_m \le K\frac{\epsilon q^2}{\lambda}. \tag{11.4}$$

Evaluating (3.1) at $q = c_0 \min(\lambda, h)$ and using Lemma 11.1 and $c_\varepsilon^2 > 3DL_0$ yields (11.2). □

12 Bootstrap iteration and proof of Theorem 1.3

Lemma 12.1 (Transfer from grouping rates to weighted variance moments)**.** *Suppose that after deleting any finite set of summands, the remaining summands can for every large J be partitioned into at most KJ groups with grouping cost $w \le KJ^{-1-\theta}$ ($\theta > 0$ fixed). Then*

$$V(h) = o(h^p) \quad \text{and} \quad W_s < \infty \qquad \text{for all } 0 < s, p < \frac{1+2\theta}{1+\theta}.$$

Proof. If $V(h) \ge ch^p$ along a sequence with $1 < p < (1+2\theta)/(1+\theta)$, replace a block of summands with $H_i \le h$ and variance $\sigma^2 \asymp h^p$ by $N(0, \sigma^2)$, group the remaining summands into $N \le KJ$ groups, and apply (9.10) with $\ell = h^\lambda$ and $J = \lceil h^{-j} \rceil$ ($\frac{p(1+\theta)}{1+2\theta} < \lambda < 1$, $\frac{\lambda}{1+\theta} < j < 2\lambda - p$) to obtain a positive objective gain $\frac{1}{2}\phi(z)\ell - o(\ell) > 0$. □

Lemma 12.2 (Initial polynomial grouping rate)**.** *Under Lemma 8.2, the hypothesis of Lemma 12.1 holds with $\theta = 1/80$, so $W_s < \infty$ for all $s < 82/81$ and $\sum_{i\ge1} \mathbb{E}|X_i| < \infty$.*

Proof. Applying Lemma 11.2 with $E_m(h) \le K$ to the shells of $\{i : v_i \le \delta\}$ above $r_\delta = \delta^{1/20}$ and using $W_{3/4} < \infty$ below r_δ gives $\sum_{v_i \le \delta} v_i/\sqrt{H_i} \le K\delta^{1/80}$, which via Lemma 9.1 yields $\theta = 1/80$. □

Lemma 12.3 (Linear envelope bound on Variance-rich shells)**.** *If $W_s < \infty$ for some $s > 1$ and a shell $h < H_i \le 2h$ contains a block of variance $\sigma^2 \ge ch^{2-\eta}$ ($\eta > 0$), then $E_m(h) \le Kh$ for every fixed $m > \max(3, 2/\eta)$.*

Proof. Applying Lemma 9.3 to the block gives $d(t) \le Kh$ for $|t| \le \sigma/\sqrt{160\log(1/h)}$, while $(h/r)^m d(u) = o(h)$ for larger r. □

Lemma 12.4 (Sublevel sums for v_i and p_iH_i)**.** *If $1 < s < 12/7$ and $W_s < \infty$, then for $q \in \{0, 1/2\}$,*

$$\sum_{v_i \le \delta} v_i H_i^{-q} \le K_s \delta^{(s-q)/6}. \tag{12.1}$$

If all sufficiently late summands are two-point variables $X_i = \varepsilon_i H_i(B_{p_i} - p_i)$ ($p_i \le 1/2$) and $1 < s < 11/6$ with $W_s < \infty$, then

$$\sum_{p_iH_i \le \delta} p_i H_i \le K_s \delta^{(s-1)/5}. \tag{12.2}$$

Proof. By Lemma 12.3 and (11.2), any shell with nonmodal intensity $\geq h^{-\eta}$ has $E_m(h) \leq Kh$, which forces $\delta \geq ch^6$ in the three-point case (and $\delta \geq ch^5$ in the two-point case). Splitting at $r_1 = L_1'\delta^{1/6}$ (or $L_1'\delta^{1/5}$) and $r_0 = L_0'\delta^{1/7}$ (or $L_0'\delta^{1/6}$) proves (12.1) and (12.2). □

Proof of Theorem 1.3 and Corollary 1.4. In the general three-point case, (12.1) with $\delta = J^{-6/(6-s)}$ gives grouping exponent $\theta_s = (2s-1)/(6-s)$, so Lemma 12.1 maps any available $s \in (1, 12/7)$ to every $\gamma < T(s) = (3s+4)/(s+5)$. Iterating T from $s > 1$ (supplied by Lemma 12.2) reaches every γ below the positive fixed point $\sqrt{5}-1$ of T. In the eventually two-point case, grouping by the sign of the nonmodal displacement has $A_- = 0$ in Lemma 9.1, so (12.2) with $\delta = J^{-5/(6-s)}$ gives $\theta_s^{\mathrm{B}} = (2s+3)/(6-s)$ and iteration map $T_{\mathrm{B}}(s) = (3s+12)/(s+9)$, whose positive fixed point is $\sqrt{21}-3$. Finally, applying (9.10) with $\ell = J^{-a}$ ($\frac{1+\alpha}{2\alpha} < a < 1+\theta$) proves Corollary 1.4. □

13 Exclusion of dense and nonoverlapping rare Bernoulli tails

In this final section we show that the variational contact equations also exclude infinite tails of rare two-point summands

$$X_j = h_j(p_j - B_j), \qquad B_j \sim \mathrm{Bern}(p_j), \qquad h_j > 0, \qquad p_j \longrightarrow 0.$$

13.1 Neumann-series probability identity at the maximizing threshold

Set $\varphi = \phi(z)$ and define the right-hand contact defect

$$d(t) = F(z) + \varphi t - At^2 + Ct^3 - F(z+t) \geq 0 \qquad (t \geq 0), \tag{13.1}$$

which by monotonicity of F satisfies

$$d(t) \leq \varphi t + |A|t^2 + Ct^3, \qquad d(y) - d(x) \leq K(y-x) \quad (0 \leq x \leq y \leq 1). \tag{13.2}$$

For a single summand $X_j = h(p-B)$ with $p < 1/2$, $q = 1-p$, and shift operator $T_h f(t) = f(t+h)$, the distribution functions $\psi(t) = F(z+t)$ and $g(t) = F_{S-X_j}(z - ph + t)$ satisfy $\psi = (qI + pT_h)g$. Inverting $qI + pT_h$ by its absolutely convergent Neumann series gives, for any polynomially bounded f with $f(0) = 0$,

$$[(T_h - I)(qI + pT_h)^{-1} f](0) = q^{-2} \sum_{k \geq 1} (-p/q)^{k-1} f(kh). \tag{13.3}$$

Since probability stationarity of X_j gives $[(T_h - I)(qI + pT_h)^{-1}\psi](0) = g(h) - g(0) = \varphi h - A(1-2p)h^2 + C(1-2p)^3 h^3$, while $(T_h - I)(qI + pT_h)^{-1}$ maps t, t^2, t^3 at 0 to $h, (1-2p)h^2, (1-6pq)h^3$, subtracting yields the exact identity

$$\sum_{k \geq 1} (-p/q)^{k-1} d(kh) = -2Cp^2q^2(3-4p)h^3 < 0. \tag{13.4}$$

In particular, moving the $k \geq 2$ terms to the right-hand side and using (13.2) gives

$$d(h) \leq \sum_{k \geq 2} (p/q)^{k-1} d(kh) \leq K_1 ph \qquad (p \leq 1/4,\ h \leq 1). \tag{13.5}$$

13.2 Exclusion of rare tails with dense gap spacings

Theorem 13.1 (Dense rare-gap exclusion)**.** *Suppose a maximizing representation contains a sequence of distinct summands $h_j(p_j - B_j)$ with $p_j \to 0$ and $h_j \downarrow 0$ satisfying, for some $\nu > 1$,*

$$h_j - h_{j+1} = O(h_j^\nu). \tag{13.6}$$

Then $F(z+t) = F(z) + \phi(z)t + O(t^{1+\alpha})$ as $t \downarrow 0$ with $\alpha = \min(1, \nu - 1)$. Consequently, under the hypotheses of Theorem 1.3, no such sequence can exist if

$$\nu > 1 + \frac{\sqrt{5}-1}{2},$$

or, when all but finitely many summands of the complete representation are two-point, if

$$\nu > 1 + \frac{\sqrt{21}-3}{6}.$$

Proof. For $1 < \nu \le 3$ and small $x > 0$, let $s(x)$ be the largest $h_j \le x$, so $0 \le x - s(x) \le K_0 x^\nu$ by (13.6). Combining (13.5) at $h = s(x)$ with (13.2) shows that $H(x) = \sup_{0 \le t \le x} d(t)$ satisfies $H(x) \le c_0 x^\nu + \sum_{k \ge 2} r_0^{k-1} H(kx)$ for small $r_0 > 0$, which by a barrier comparison with $K_2(x^\nu + x^3)$ implies $d(t) = O(t^\nu)$ as $t \downarrow 0$. Evaluating the phase-rounded array (9.10) at $z + b\ell$ ($b \asymp \sqrt{J \log J}$, $\ell = J^{-a}$ with $\frac{1+\alpha}{2\alpha} < a < 1 + \theta$) then contradicts maximality whenever $\alpha > 1/(1+2\theta)$, which under the grouping rates $\theta = (2s-1)/(6-s)$ ($s < \sqrt{5} - 1$) and $\theta = (2s+3)/(6-s)$ ($s < \sqrt{21} - 3$) of Theorem 1.3 yields the stated thresholds. □

13.3 Exclusion of nonoverlapping rare tails

Theorem 13.2 (Nonoverlapping rare-tail exclusion)**.** *No maximizing representation can have the form*

$$S = P + \sum_{j \ge 1} h_j(p_j - B_j),$$

where P has finite support and is independent of the tail, $0 < p_j \le 1/2$ with $p_j \to 0$, and for some constant $c > 0$,

$$T_j := \sum_{k > j} h_k \le h_j, \qquad h_{j+1} \ge c h_j > 0$$

for all sufficiently large j (for example, $h_j = a r^j$ with $0 < r \le 1/2$ and $p_j \to 0$).

Proof. Put $M(t) = F(z+t) - F(z)$. By (13.1)–(13.5) and $h_j \ge h_{j+1} + h_{j+2} \ge 2h_{j+2}$, we have $M(t) \le K_3 t$ and

$$M(h_j) - M(h_{j+2}) \ge \frac{\varphi h_j}{3} \tag{13.7}$$

for all large j. For each atom $a \in \operatorname{supp} P$ of mass w_a with $y_a = a + \sum_j p_j h_j - z = \sum_j h_j \varepsilon_{a,j} \in \{\sum_j h_j \varepsilon_j\}$, the prefix weight $v_{a,j} = w_a \prod_{k<j} p_k^{\varepsilon_{a,k}} (1-p_k)^{1-\varepsilon_{a,k}}$ at any index with $\varepsilon_{a,j} = 1$ satisfies

$$v_{a,j}(1 - p_j) \le M(2h_j) \le 2K_3 h_j, \qquad v_{a,j} \le 4K_3 h_j. \tag{13.8}$$

Because $T_j \le h_j$ and $p_j \to 0$, contributions to $M(h_i) - M(h_{i+2})$ from first-difference digits $j < i$ are $o(h_i)$, while digits $> i+2$ have displacement $\le T_{i+2} \le h_{i+2}$. Thus for each large i, some $a \in \operatorname{supp} P$ and $j \in \{i, i+1, i+2\}$ must have $\varepsilon_{a,j} = 1$ and $v_{a,j} \ge [\varphi/(12|\operatorname{supp} P|)]h_j$, which forces $v_{a,k}/h_k \le 4K_3 c^{-(k-j)} p_j \to 0$ for the next $L = 3|\operatorname{supp} P| + 3$ indices, contradicting the pigeonhole principle over $|\operatorname{supp} P| + 1$ disjoint three-index windows. □

A Exact rational bounds for Lemma 5.3

For $x \geq 0$, the Taylor polynomials $E_n(x) = \sum_{k=0}^{n} x^k/k!$ and $U_n(x) = E_n(x) + \frac{x^{n+1}}{(n+1)!(1-x/(n+2))}$ $(0 \leq x < n+2)$ bracket $E_n(x) \leq e^x \leq U_n(x)$. With step $d = 3/100$, $l_j = jd$, and $r_j = (j+1)d$, the upper Riemann sum

$$I \leq \sum_{j=0}^{199} \frac{d\, r_j^2}{E_{40}(l_j^2/2 - (269/5000) l_j^3)} < \frac{23}{10} \tag{A.1}$$

and $J \leq \frac{6025}{216 E_{40}(108/25)} < \frac{3}{8}$, together with $U_{24}(25/54)/[6(157/50)] < 17/200$ and $11/[19 E_{40}(361/289)] < 1/6$, are verified in exact rational arithmetic by the companion script `gaussian_window_certificate.py`, yielding

$$\frac{\Delta(S)}{\beta} < \frac{1182125003}{3000000000} < \frac{2}{5}.$$

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